EDM cutting plans

User guide of the cutting-plan tool (`cutting-sheets/`) – version of 10 November 2026

1. What the tool does

You write a short order (a text file: which specimens, how many, at which angle to the rolling direction, on which sheet). The tool then, without any manual step:

Step	What happens
1. Order and templates	reads and checks the order; takes the template of each specimen type from the lab catalogue
2. Specimens	turns each specimen to its angle, adds its notch code (angle + index) and its 2 Mikrosteg recesses
3. Placement	places all the specimens on as few sheets as possible: 5 mm gap, inside the effective area of the machine, never mirrored
4. DXF	writes the cutting plan (mm, colours = EDM quality Q3 / Q4 / Q5)
5. Checks	reads the final DXF back and checks contours, gaps, margins, outlines and codes
6. Notice and registry	writes the notice for the workshop and the specimen registry

Typical durations: shear plan (24 specimens) about 30 s; plans of 64–66 specimens about 4 min.

2. Installation (once per PC)

The tool runs on Windows with Python 3.10 to 3.14 (64 bit) and Git Bash. Open Git Bash and check the Python version:

```
py --version
```

Then install the Python modules:

```
py -m pip install ezdxf shapely matplotlib scipy pyyaml spyrrow==0.9.0 packingsolver
```

Get the repository the first time (then `git pull` is enough to update it):

```
cd ~/Documents
git clone https://github.com/josselindavion/eth.git
```

> If `py` starts an old Python (for example 3.7, 32 bit), spyrrow cannot be installed: install a recent Python 3 (64 bit) from python.org and run `py -0p` to list the installed versions.

3. Running an order

Always from the repository root:

```
cd ~/Documents/eth
git pull
py cutting-sheets/scripts/make_plan.py cutting-sheets/orders/CAMPAIGN_600x500mm_WATERJET.yaml
```

Option	Effect
(none)	results in `cutting-sheets/runs/<name>/`; this folder is ignored by git, so runs can be repeated freely
`--final`	results in `cutting-sheets/output/<name>/` (kept by git): use it for the plan sent to the workshop, then `git add`, `git commit`, `git push`
`--time 5`	replaces `time` of the order (seconds of compaction per sheet)

The terminal shows the 6 steps, progress bars with the remaining time, the specimens moved between sheets, and the table of checks. A good run ends with:

```
DONE, all checks OK, in 2 min 08 s
    Plan    : .../runs/CAMPAIGN_600x500mm_WATERJET/CAMPAIGN_600x500mm_WATERJET.dxf
    Notice  : .../notice.pdf
    Registry: .../registry.csv
    Preview : .../preview.png
```

4. Writing an order

Copy `cutting-sheets/orders/CAMPAIGN_600x500mm_WATERJET.yaml` under a new name and edit it with any text editor. Every field is explained by a comment in the file. Text after # is a comment. Keep the indentation (spaces, no tabs).

```
name: ETH_300x400mm_SHEET_v20261103_UT30   # plan name: output folder, DXF, notice
sheet: 300x400                      # complete sheet, RD x TD (rolling direction first)
clamping: {left: 40, bottom: 40}    # L-shaped clamping
machine_area: 250x350               # wire travel of the EDM (CUT E 350), RD x TD
placement_margin: 5                 # tolerance on the position of the sheet in the machine
edge_margin: 5                      # distance to a free edge of the sheet
gap: 5                              # distance between two specimens
specimens:
  - {type: UT40, angle: 30, count: 5}
  - {type: NT20, angle: 45, count: 3, first_index: 1}
```

4.1 The sheet on the machine

Field	Meaning	Default
sheet	complete sheet in mm, rolling direction first: RDxTD (600x500 = 600 mm along the rolling direction × 500 mm across)	required
clamping	bands held by the clamps, per side (left, bottom, right, top); {left: 40, bottom: 40} = L-shaped clamping	none
machine_area	area the wire can reach, RD × TD (CUT E 350: 250x350 = Y × X travel); leave it out for the waterjet	whole sheet
machine_position	where the machine area starts on the sheet, {left: .., bottom: ..}	inner corner of the clamping
placement_margin	nothing is placed within this distance of the machine area edge, because the sheet is not positioned to the mm	5
edge_margin	minimum distance to a free (not clamped) edge of the sheet	5
gap	minimum distance between two specimens	5

Effective area (where the specimens go) = machine area minus the placement margin on each side, inside the sheet and outside the clamping. With the values above: machine area from x 40, y 40 to x 390, y 290, effective area 240 (RD) × 340 (TD) mm from x 45, y 45. The terminal, the notice and the preview show these zones and write every size rolling direction first, marked (RD) / (TD).

4.2 Placement time

Field	Meaning	Default
assign_time	seconds for PackingSolver to put all the specimens on sheets	30
fill_tries	failed tries allowed when moving specimens to empty one sheet (0 = faster, maybe one sheet more)	8
fill_time	seconds at most per try (a try that succeeds stops after 1–3 s)	10
time	seconds at most of final compaction per sheet (0 = none)	15
seed	random seed: change it to get another layout	1
max_sheets	the order is rejected if it needs more sheets	20
spread_distance	two repeats (same type and angle) closer than this (mm) are "side by side": swaps after the placement separate them; the pairs left are a warning (orange circles on the preview)	10
spread_tries	placements tried (seed, seed + 1, ...) to keep the repeats apart; the best is kept (fewest sheets, then fewest pairs); time up to x tries	1

4.3 Specimens

One line = count specimens of one type at one angle.

Key	Meaning
type	specimen type (table below)

Key	Meaning
angle	direction of the coded end from the rolling direction, counter-clockwise: 0 to 135° in steps of 15° (written in the notch code)
count	number of specimens
first_index	index (number in the notch code) of the first specimen, then +1 each (default 0, max 9)

Type (lab name)	Older names	Specimen
UT40	UTUT, UT	uniaxial tension, 106 mm (gauge zone ≈ 40 mm)
NT20 / NT06	NT6.67 (for NT06)	notched tension, radius 20 / 6.67 mm, 106 mm
CHD48	CHCH, CH	central hole Ø 4.8 mm, 106 mm
UT19	UTST	short uniaxial tension, 81 mm (gauge zone ≈ 19 mm)
SHLD / SHMD / SHHD	–	short shear, 81 mm
MNAK_ROUND_FULL / MNAK_ROUND_HOLE	MNAK_FULL / MNAK_HOLE	round Mini-Nakajima without / with 2 holes (holes at angle + 90°)
MNAK_RECT_FULL / MNAK_RECT_HOLE	– / MNAK_RECT	rectangular Mini-Nakajima 80 × 54 mm without / with 2 holes (holes oriented as on the disc with the same angle)
LASER	TENS_FULL, STRIP	strip 140 × 30 mm, no code (speckle contrast)

Only the fields of this guide are accepted: an unknown field (typo, old field) stops the run with the list of allowed fields, so nothing is silently ignored.

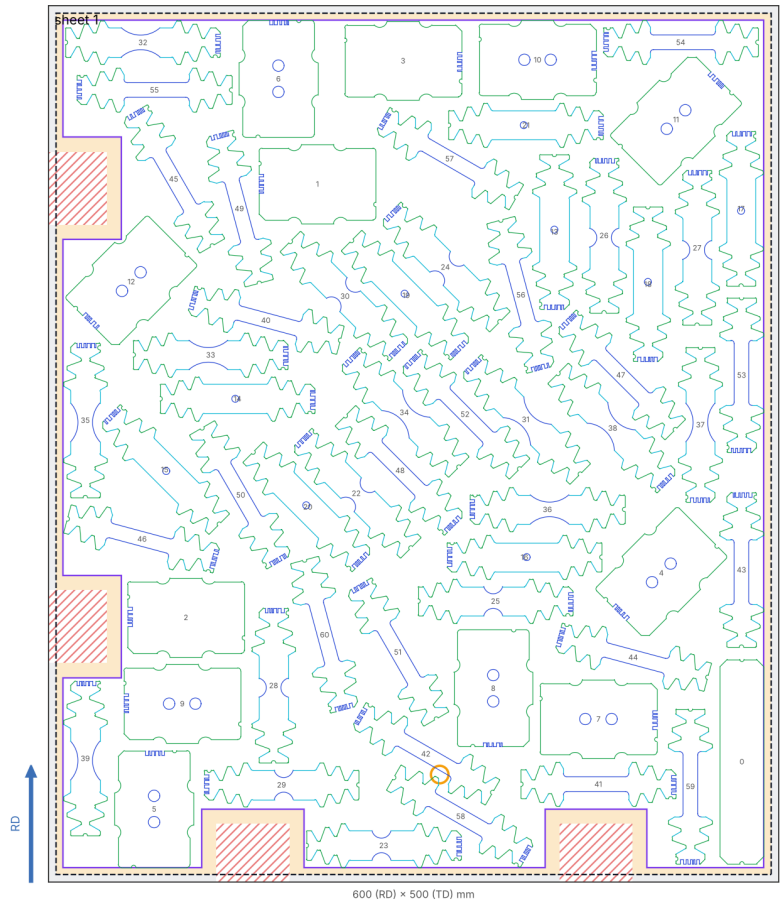
5. Results

Each run writes exactly 4 files in its folder:

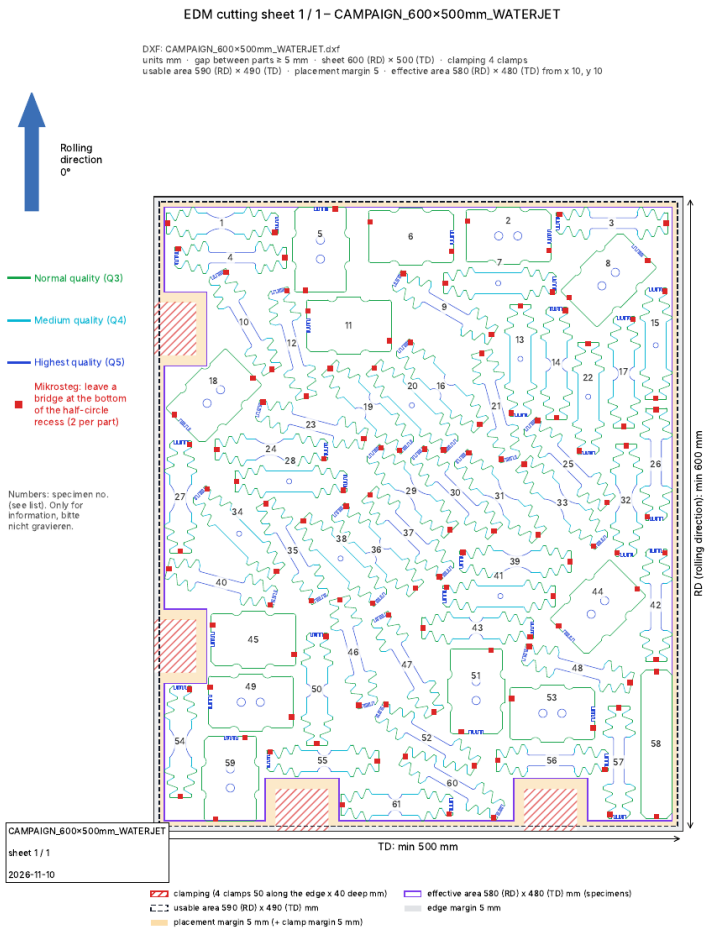
File	For	Content
<name>.dxf	workshop	the cutting plan: one frame per sheet, every specimen with its notch code; mm; colour = EDM quality (3 = Q3 normal, 4 = Q4 medium, 5 = Q5 high)
notice.pdf	workshop	one page per sheet (numbered specimens, rolling direction, qualities, zones, Mikrostegs as red squares) and the list of specimens
registry.csv	lab	one line per specimen: number, sheet, type, angle, notch code, position, specimen ID
preview.png	you	all the sheets with the zones, to check the plan at a glance

Mikrostegs: every specimen has 2 small half-circle recesses (radius 1.5 mm) in its outline. The workshop stops the wire at the bottom of each recess (red square on the notice), so the specimen stays attached; after breaking it out, the rest of the bridge stays inside the recess.

Send the DXF and notice.pdf to the workshop; keep registry.csv for the lab.



preview.png of the 600 × 500 campaign (orders/CAMPAIGN_600x500mm_WATERJET.yaml). Red hatching: clamps; black dashed: usable area; orange band: placement margin; purple: effective area where the specimens are placed; orange circle: repeats side by side; blue arrow: rolling direction.



First page of notice.pdf (same plan); red squares = Mikrostegs.

6. Checks and error messages

After writing the DXF, the tool reads it back without any assumption. Every line must be 0K:

Check	What is verified
Units	the DXF is in millimetres
Sheets	one frame per sheet, of the ordered size
Closed contours / duplicate lines	every cutting contour is closed, no line twice
Colours = EDM quality	cutting lines only in colours 3 / 4 / 5
Overlaps / gap between pieces	no overlap; at least gap mm between two pieces
Margin to sheet edge	every piece inside the effective area
ID codes	every coded specimen carries a notch code; shared codes only between different shapes
Specimens vs order	outline = template, code read back = code ordered, direction = angle ordered

Message	Meaning and what to do
ORDER REJECTED: unknown field(s): …	typo or old field in the order: use the names of section 4
ORDER REJECTED: missing field "machine_area"	sheet and machine_area are required
ORDER REJECTED: no effective area left	clamping + margins leave no room: check sheet, clamping, machine_area, placement_margin
ORDER REJECTED: … does not fit in the usable area	a specimen is larger than the effective area once turned: bigger sheet or smaller margins
ORDER REJECTED: the specimens need N sheets	more sheets than max_sheets: raise max_sheets or split the order

Message	Meaning and what to do
DONE WITH ERRORS (see above)	a check failed: do not send the plan; the preview circles the problem in red
py314: command not found	use py (the alias py314 only existed in one terminal)

7. Tips

I want…	Do this
a faster run	lower fill_tries (0 = no try to remove a sheet) and time
fewer sheets	raise fill_tries / fill_time, or try another seed
a different sheet or clamping	change sheet, clamping, placement_margin: the preview shows the new zones
to keep the plan sent to the workshop	run again with --final, then git add / commit / push

More details (method, engines, notch code): `cutting-sheets/README.md`.